Biofilm Susceptibility to Combination Treatment of Brazilin and APDT

Background

Biofilms: Dense bacterial communities, protected against environment by extracellular polymeric substances (EPS)

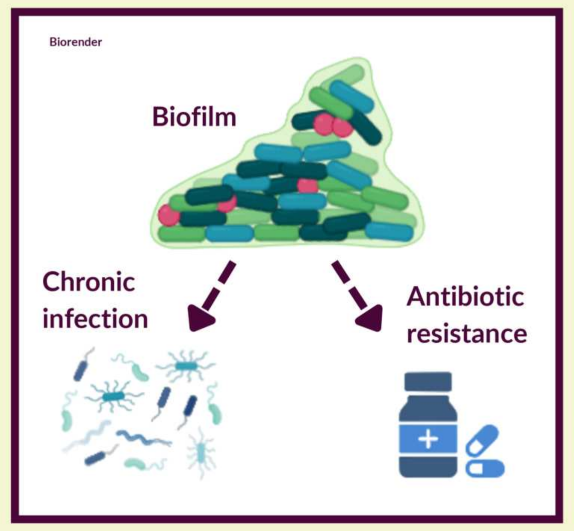


Fig. 1: pertinent characteristics of bacterial biofilms

Brazilin

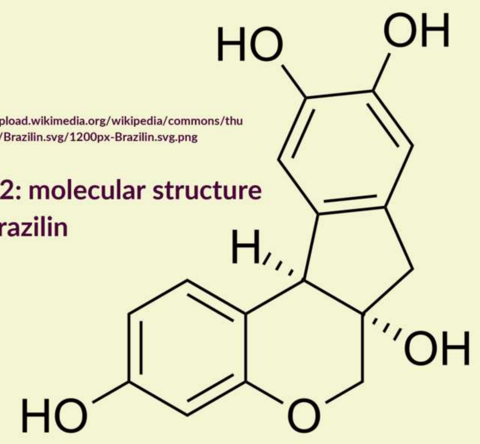


Fig. 2: molecular structure of brazilin

- Natural dying agent
- Antimicrobial properties
- Treated antibiotic resistant *S. aureus* biofilms combined with vancomycin

Photodynamic Therapy

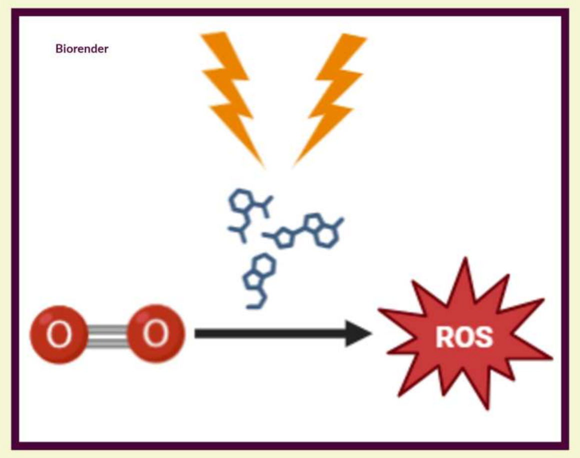


Fig. 3: mechanism of action of APDT

Methylene blue mediated photodynamic therapy (APDT)

- Effect at disrupting EPS protection
- Strong synergetic effect with antibiotics

Purpose

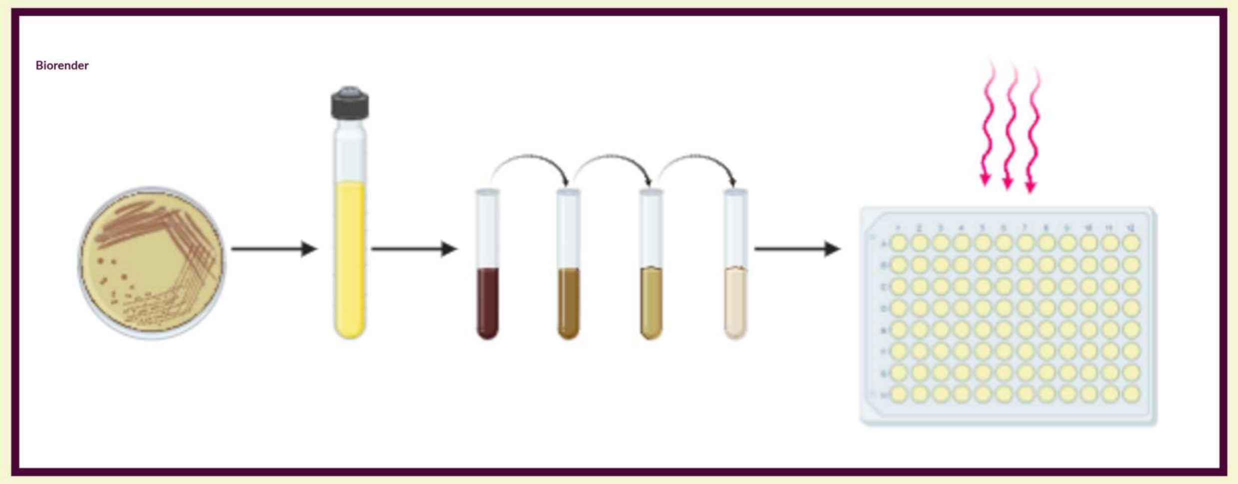
Purpose: To determine the combined effect of brazilin and methylene blue mediated photodynamic therapy on inhibiting *P. fluroescens* biofilms.

Hypothesis: Biofilms treated with photodynamic therapy followed by brazilin will be greatly inhibited because photodynamic therapy disrupts the outer EPS layer, enabling brazilin to target the vunerable inner layers of the biofilm.

IV: concentration of brazilin (0, 16, 32 µg/mL) and methylene blue (0, 25 µg/mL)
DV: biofilm survival percentage measured by crystal violet staining

Methodology

Culturing *P. fluorescens* biofilms



1. Streak *P. fluorescens* on agar plate with 4 quadrant method
2. Inoculate colonies to culture tube following incubation
3. Following culture growth, read absorbance with spectrophotometer at 600 nm and dilute to 0.01 absorbance
4. Transfer bacteria and media to 96 well plates
5. Apply 580 nm irradiation for 15 minutes
6. Incubate plates for 48 hours

Fig. 4: biofilm culturing protocol

Crystal violet staining

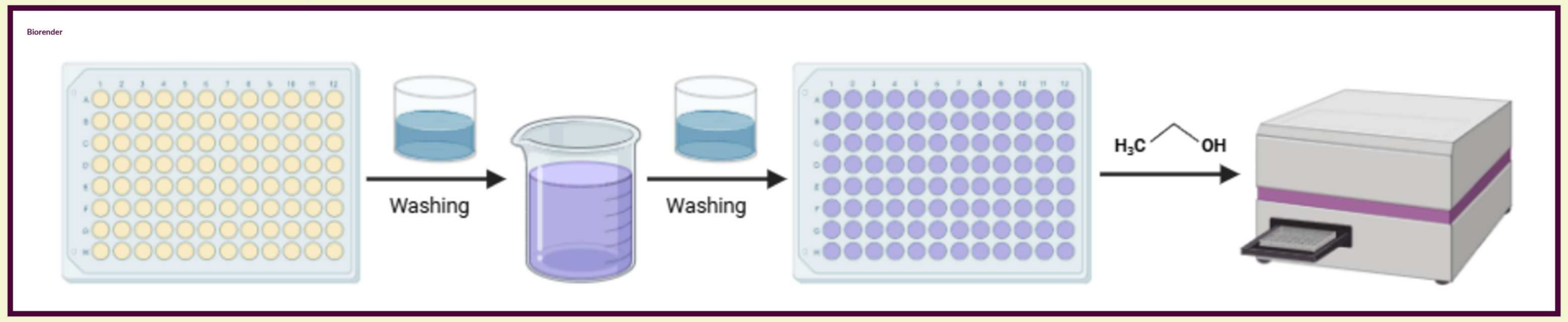
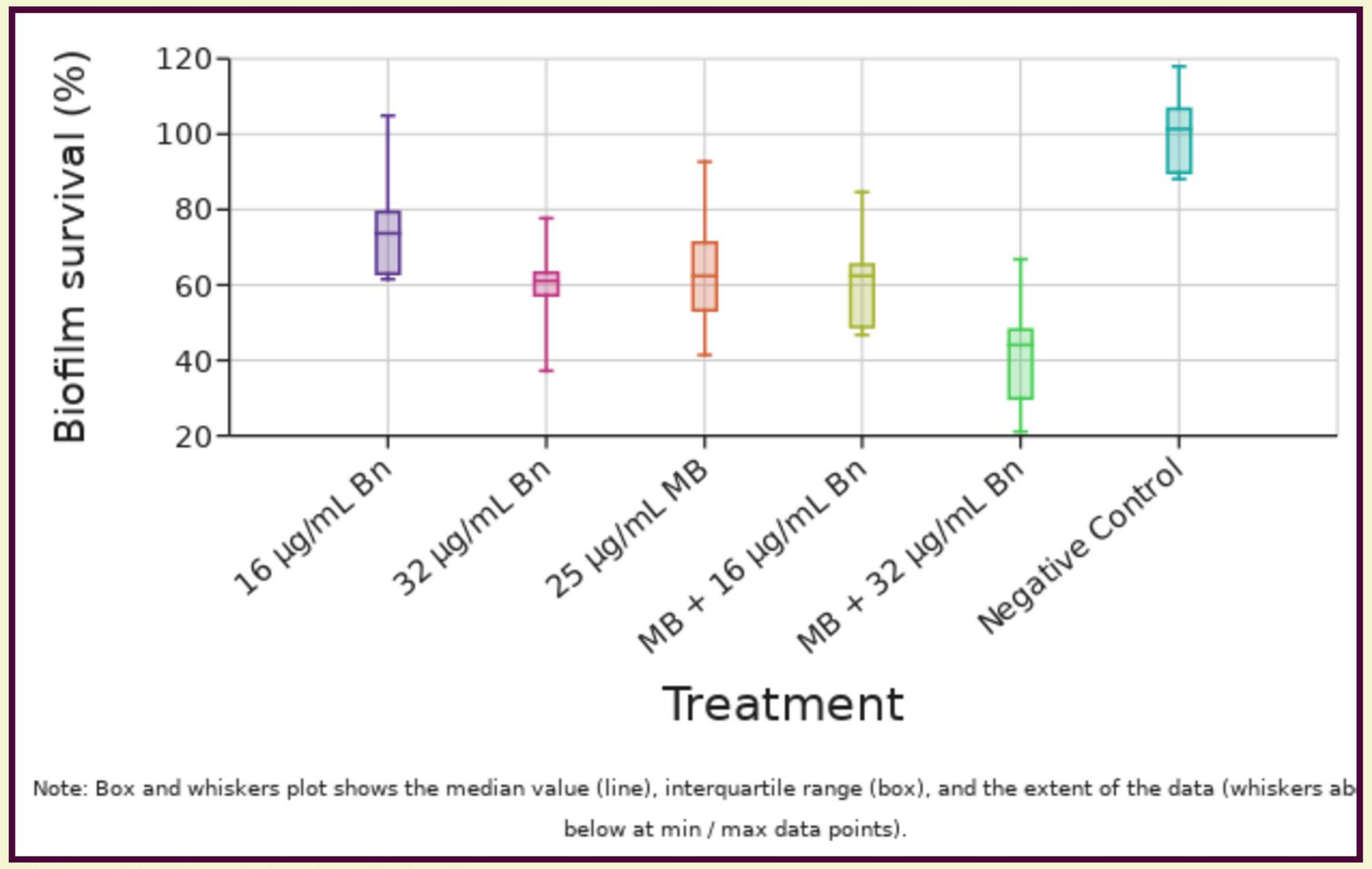


Fig. 5: crystal violet staining and solubilization protocol

1. Following incubation, remove media
2. Wash in a water bath to remove any non-biofilm cells
3. Stain wells with crystal violet solution and incubate at room temperature for 15 minutes
4. Repeat washing three times to remove any non-adherent crystal violet
5. Dry plates at room temperature for 30 minutes
6. Solubilize crystal violet in ethanol
7. Read absorbance of crystal violet at 570 nm with a plate reader

Results

Fig. 6: Biofilm survival percentages of biofilm wells following treatment and staining



MB=25 µg/mL MB=methylene blue; Bn=brazilin

Fig. 7: Well plates stained wtih crystal violet

Fig. 8: Preliminary APDT setup

Discussion

- The data was analyzed with a Kruskal Wallis and Dunn's post hoc nonparametric test, and the results partially supported the hypothesis. KRW returned a significant $p<0.01$, H-stat=51.69 at df=5
- At all concentrations except for 16 µg/mL biofilm survival was reduced compared to no treatment ($p<0.01$)
- The greatest reduction in survival was reported in treatment with 32 µg/mL Bn + 25 µg/mL MB, leading to a median 55.9% reudction
- However, the only significant difference in survival across treatments was between 16 µg/mL Bn and 32 µg/mL Bn + 25 µg/mL MB ($p<0.01$)
- Low treatment concentrations could have induced biofilm growth as a response to environmental stress
- Concentrations could have also been too low to meaningfully impact biofilm formation of *P. fluorescens*, since no prior study has used brazilin or methylene blue mediated photodynamic therapy to treat *P. fluorescens* biofilms

Future Work

- Employing different concentrations of brazilin and methylene blue
- Treatment against dual species biofilms (ex. *P. fluorescens* and *L. monocytogenes*)
- Analysis of changes in biofilm morphology using scanning electron microscopy
- Analysis of changes in biofilm forming factors such as quorum sensing and gene expression

References

1. Huang, T.C., Chen, C.J., Ding, S.J., & Chen, C.C. (2019). Antimicrobial efficacy of methylene blue-mediated photodynamic therapy on titanium alloy surfaces in vitro, Photodiagnosis and Photodynamic Therapy, 25, 7-16. DOI: https://doi.org/10.1016/j.pdpdt.2018.11.008
2. Limoli, D.H., Jones, C.J., & Wozniak D.J. (2014). Bacterial extracellular polysaccharides in biofilm formation and function, Microbiology Spectrum, 3(3). DOI: https://doi.org/10.1128/microbiolspec.mb-0011-2014
3. Pattananandecha, T., Apichai, S., Julsrigival, J., Ogata, F., Kawasaki, N., & Saenjum, C. (2022). Antibacterial activity against foodborne pathogens and inhibitory effect on anti-inflammatory mediators' production of brazilin-enriched extract from Caesalpinia sappan (Linn.), Plants, 11(13), 1698. DOI: https://doi.org/10.3390/plants11131698
4. Songca, S.P. & Adjei, Y. (2022). Applications of antimicrobial photodynamic therapy against bacterial biofilms. International Journal of Molecular Sciences, 23(6), 3209. DOI: https://doi.org/10.3390/ijms23063209